

Octal to Decimal Conversion

$r = 8$

I.P. F.P.

$O_3 \ O_2 \ O_1 \ O_0 . O_{-1} \ O_{-2}$

↑
Octal point

$$O_3 \times 8^3 + O_2 \times 8^2 + O_1 \times 8^1 + O_0 \times 8^0 + O_{-1} \times 8^{-1} + O_{-2} \times 8^{-2} = (\quad)_{10}$$



$$\begin{array}{c}
 \swarrow \quad \swarrow \quad \swarrow \quad \swarrow \quad | \quad \swarrow \quad \swarrow \\
 0_3 \times 8^3 + 0_2 \times 8^2 + 0_1 \times 8^1 + 0_0 \times 8^0 + 0_{-1} \times 8^{-1} + 0_{-2} \times 8^{-2} = (\quad)_{10}
 \end{array}$$

Octal point

Ex: - i) $(57.4)_8 \longrightarrow (47.5)_{10}$

ii) $(\underline{4507.44})_8 \longrightarrow (\quad)_{10}$

$$5 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1}$$

$$40 + 7 + \frac{4}{8}$$

$$47 + 0.5$$

$$\underline{47.5} \text{ Ans}$$

$$4 \times 8^3 + 5 \times 8^2 + 0 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1} + 4 \times 8^{-2}$$

$$4 \times 512 + 5 \times 64 + 0 \times 8 + 7 \times 1 + \frac{4}{8} + \frac{4}{16}$$

$$2048 + 320 + 0 + 7 + \frac{8+4}{16}$$

$$2375 + 0.5625$$

$$= (\underline{2375.5625})_{10}$$



Ex:- i) $(57.4)_8 \longrightarrow (47.5)_{10}$

$$5 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1}$$

$$40 + 7 + \frac{4}{8}$$

$$47 + 0.5$$

$$47.5 \text{ Ans}$$

ii) $(\underline{4507.44})_8 \longrightarrow ()_{10}$

$$4 \times 8^3 + 5 \times 8^2 + 0 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1} + 4 \times 8^{-2}$$

$$4 \times 512 + 5 \times 64 + 0 \times 8 + 7 \times 1 + \frac{4}{8} + \frac{4}{64}$$

$$2048 + 320 + 0 + 7 + \frac{8+1}{16}$$

$$2375 + 0.5625$$

$$= (2375.5625)_{10}$$

1.385 i) $(601)_8 \longrightarrow ()_{10}$

ii) $(1070.02)_8 \longrightarrow ()_{10}$



1. 385

2. 568.0312